

Abstract

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Investigating Accountability in Business Systems-of-Systems

Felipe Cordeiro

Instituto de Tecnologia de Support Technical Education - FAETEC
Rio de Janeiro, RJ, Brazil
felipecordeiro@faetec.rj.br

Rodrigo Pereira dos Santos

Universidade do Estado do Rio de Janeiro - UNIRIO
Rio de Janeiro, RJ, Brazil
rps@uniriotec.br

Aline Viana

Fluminense Federal University
Campos dos Goytacazes, RJ, Brazil
apireis@fluminense.br

Patrícia

Vrije Universiteit Amsterdam
Amsterdam, The Netherlands
p.la@vu.nl

The evolution of business processes has driven the systems-of-systems (SoS) across various domains, including technologies such as cloud computing, e-commerce platforms, and smart environments. In this context, the integration of heterogeneous and independent constituent systems into a unified system that collaborates to achieve business goals and accountability of these systems must be a concern, as current accountability approaches can obscure the responsibility of data, processes, and outcomes. **PROBLEM:** The current state of the art often results in studies offering specific solutions, neglecting the ongoing need for a shared understanding of the problem. Furthermore, establishing accountability as a quality attribute is a significant challenge due to limited research and a fragmented agenda for underlying challenges. **SOLUTION:** This paper presents an overview of accountability from reporting requirements to proposing a research agenda to address these challenges. **METHOD:** The study adopts a prescriptive approach through a systematic mapping study. **RESULTS:** The study identifies key aspects of accountability, and a research agenda when considering various topics for further investigation. **CONCLUSION:** Based on the knowledge on accountability, this study facilitates the expansion of the body of knowledge on the field and brings forth innovative solutions.

PTS

Software engineering → Software creation and man-

agement complexity differs from traditional operational and managerial systems, and evolutionary behaviors [6]. In this context, SoS emerges: system-of-independent systems, including their constituent systems, and management for supporting them.

SoIS exhibit a strong behavior expressed as dynamic and interactive [9]. While SoS primarily focus on hardware components, SoIS take on various elements within the system by exhibiting unique properties. Unlike business-process-oriented systems, from interoperability among traditional systems, a SoIS involves connections, potentially source code, developers, organizations), and regulatory frameworks. The importance of considering not only the system but also the context.

Despite many benefits, SoIS, particularly in terms of accountability [12]. *Accountability is a non-judicial responsible actions in organizations [14] and must encompass the obligation of individuals for their actions, decisions, and activities [16].* For example, in the case of services, or a security breach, it becomes intricate due to its